a)

$-m\tau_n^z$

G_n

$-g\sigma^z_{(n,v)}$

$-\lambda\tau^x_{n+v}\sigma^x_{(n,v)}\tau^x_n$

b)

g

$(5, 2)$

Confined

$(0.3, 0.5)$

Higgs

$(5, 0.01)$

Deconfined

m

Energy gap

2.5

1.5

0.5

0.2

0.1

0.0

0

1

2

3

c)

Confined
Yoyo

Deconfined
Spreading

Higgs
Glassy

Confined
Bending

d)

$\mathcal{C}$ $\mathcal{C}^{-1}$ $\mathcal{C}$ $\mathcal{C}^{-1}$

e)

$\exp(-ig\sigma^z_{(n,v)}dt/4)$

$\exp(-im\tau^z_n dt/4)$

$\exp(-i\lambda\sigma^x_{(n,v)}dt/2)$

$\exp(-i\tau^z_n\phi_{k,n})$